CLASS 8 — MATHS · FORMULA SHEET

Chapter 12: **Tales by Dots and Lines** | Quick revision, one page

Measures of central tendency

- Mean $= \dfrac{\text{sum of observations}}{\text{number of observations}}$.
- Median: middle value of the data arranged in order. Odd n: the $\dfrac{n+1}{2}$th value. Even n: mean of the $\dfrac{n}{2}$th and $\left(\dfrac{n}{2}+1\right)$th values.
- Mode: the observation that occurs most often (a data set may have more than one mode, or none).

Spread

- Range $=$ largest observation $-$ smallest observation.

Useful results

- Sum of observations $=$ mean $\times$ number of observations.
- Mean lies between the smallest and the largest observation.
- Adding a value equal to the mean does not change the mean.

Representing data

- Dot plot: one dot for each occurrence of a value above a number line; the tallest column marks the mode.
- Line graph: points joined by lines; used to show change over time (rise, fall, trend).
- Read a line graph by comparing values at consecutive points; change $=$ later value $-$ earlier value.

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